



राष्ट्रीय प्रौद्योगिकी संस्थान नागालैंड

NATIONAL INSTITUTE OF TECHNOLOGY NAGALAND

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TEXMiN - MINING CPS CENTRE OF EXCELLENCE

UG/PG Fellowship - Project Closure Report

1. Title of the Project

Edge-Enabled Smart Mine Guardian System: A CPS Framework for IIoT-Based Safety and Monitoring

2. Project Reference Number

TEXMiN-CoE/UG-PG-Fellowship/2025/09-005

3. Total Duration of Project

10 Months

4. Name, Registration No., Department & Year of Fellow

Name: Peniel Chang

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Department: Electronics and Instrumentation Engineering

Year: B.Tech Final Year

5. Name, Designation & Institution/Organization of Faculty Mentor

Mentor: Dr. Daijiry Narzary

Designation: Assistant Professor

Department: Electronics and Instrumentation Engineering

Institution: National Institute of Technology Nagaland

6. Project Objectives

The primary objective of this research project is to conceptualize, design, and develop a cost-effective, edge-enabled Cyber-Physical System (CPS) aimed at enhancing the safety of underground mining environments through the deployment of Industrial Internet of Things (IIoT) technologies. The specific objectives outlined for this endeavor include:

- Designing and implementing an edge-enabled multi-sensor monitoring infrastructure tailored for subterranean mining environments.
- Enabling the continuous, real-time monitoring of critical environmental parameters, specifically methane gas, carbon monoxide, temperature, humidity, and structural vibrations.
- Deploying advanced machine learning algorithms, notably Isolation Forest and Random Forest, to facilitate robust and autonomous anomaly detection.
- Developing an AI-powered computer vision module leveraging the YOLO architecture for the automated detection of Personal Protective Equipment (PPE) compliance among workers across multiple camera feeds, including entrance gates and various subterranean tunnels (e.g., Tunnel 1).
- Implementing an RFID-based localization and identification system to track mine personnel efficiently.
- Establishing a resilient communication framework utilizing LoRa and MQTT protocols to ensure reliable data transmission in inherently low-connectivity mining settings.
- Designing and deploying a portable, LoRa-enabled multi-modal alert device for mine workers, equipped with OLED visual warnings, auditory buzzers, haptic vibration alerts, and an integrated SOS emergency response button.
- Creating an intuitive web and mobile-accessible dashboard, alongside a dedicated supervisor application, for real-time data visualization and instantaneous emergency alerting.

7. Deliverables Achieved Till Date

To date, the project has successfully met several key deliverables, establishing a solid foundation for the subsequent implementation and testing phases:

- A comprehensive literature survey focusing on IIoT, CPS, edge computing, and AI-driven mine safety systems has been successfully concluded.
- The overarching system architecture has been thoroughly designed and successfully implemented.
- Environmental sensing modules integrating DHT22 (temperature/humidity), MQ4 (methane), MQ7 (carbon monoxide), and vibration sensors have been fully developed and calibrated.
- Edge processing units utilizing ESP32 microcontrollers and Raspberry Pi have been seamlessly integrated into the system framework.
- Machine learning models (Isolation Forest and Random Forest) have been deployed specifically for environmental anomaly detection at the edge.
- A functional YOLO-based PPE detection module capable of processing multiple camera streams simultaneously for comprehensive spatial coverage has been developed, trained, and tested.
- Development of a portable ESP32-based worker device integrating an SX1278 LoRa module, 0.96-inch OLED display, buzzer, vibration motor, and an SOS push button for bidirectional emergency communication.
- Development of a dedicated supervisor application for remote monitoring and coordination.
- An RFID-based worker identification and access monitoring system has been successfully established.
- A real-time monitoring dashboard, equipped with automated hazard alerting capabilities, has been developed and is fully functional.
- The LoRa and MQTT hybrid communication framework has been successfully integrated and evaluated for data transmission reliability.
- A functional laboratory-scale prototype has been demonstrated, coupled with successful initial validation.

8. Intellectual Property Rights (IPR) Achievements

- Patent: Currently in the developmental stage; filing is planned for the near future following comprehensive field validation.
- Copyright: Nil.
- Publications: A comprehensive research manuscript detailing the proposed methodology and preliminary findings is currently under preparation for submission to a peer-reviewed scientific journal.
- Software & AI Models: Proprietary embedded software and custom-trained AI models have been successfully developed as integral components of this project.

9. Progress & Accomplishments

Significant and measurable progress has been achieved during the current reporting period. The complete hardware architecture has been meticulously designed and successfully integrated with a specialized suite of environmental sensors. Edge computing nodes, leveraging the processing capabilities of ESP32 and Raspberry Pi, have been configured to handle local data processing efficiently, thereby drastically reducing system latency.

In terms of algorithmic development, machine learning models based on Isolation Forest and Random Forest have been trained, optimized, and evaluated for the precise detection of environmental anomalies. Concurrently, the YOLO-based computer vision system has been expanded to support multi-camera inputs, enabling continuous PPE compliance monitoring not only at primary access points but extending deep into operational zones such as Tunnel 1. Furthermore, a robust RFID-based system for the continuous identification and tracking of workers has been fully implemented.

Significant progress was also made in developing localized hardware solutions. We successfully engineered a portable, battery-operated (18650 cell) device for individual workers. Driven by an ESP32 microcontroller and a LoRa module (SX1278/RA-02), the device serves as a vital bidirectional communication link. Upon receiving control room alerts, it initiates a multi-modal warning—triggering an OLED message, an auditory buzzer, and a haptic vibration motor via a MOSFET driver—ensuring workers are

notified even in high-noise industrial environments. Conversely, an integrated SOS push-button allows workers to instantly transmit emergency distress signals back to the control room, with visual confirmation ("SOS Sent") displayed on the OLED screen. Additionally, a dedicated supervisor application was developed to streamline alert management and oversee field operations.

The comprehensive graphical monitoring dashboard has been significantly refined to offer granular, real-time insights, as evidenced by recent system validations. The interface now successfully visualizes localized environmental telemetry (e.g., specific parameters for Tunnel 1) alongside contextual AI-driven diagnostic analysis, such as autonomously flagging hazards like "High Humidity". Furthermore, the dashboard features a robust access control and worker tracking system that logs precise personnel locations and "Last Seen" timestamps. It seamlessly interfaces with the multi-camera PPE detection module; for instance, if a worker attempts entry without the requisite helmet or vest, the system immediately registers an "ACCESS DENIED" state and meticulously logs the infraction in a continuous Activity Feed. Real-world visual validations of the YOLO model confirm highly accurate, real-time bounding box detection of hard hats and safety vests with strong confidence scores.

The underlying communication infrastructure, which utilizes a hybrid combination of LoRa, Wi-Fi, and MQTT protocols, has undergone rigorous testing. It has proven highly reliable for continuous data transmission, even within simulated low-connectivity environments.

10. Milestones Achieved

Milestone 1 (Achieved in Sep-Oct 2025): Completed a comprehensive literature review, finalized the system architecture design, developed an initial PPE verification prototype using YOLO, and successfully integrated early hardware (ESP8266 & DHT22) to validate the Isolation Forest model for anomaly detection.

Milestone 2 (Achieved in Nov-Jan 2026): Implemented the multi-sensor environmental monitoring system (MQ-7, MQ-5, vibration sensors) and deployed the foundational edge

computing framework utilizing ESP32. Achieved successful sensor data fusion (methane & temperature) to drastically reduce false alarms.

Milestone 3 (Achieved in Feb-Apr 2026): Successfully integrated and deployed advanced machine learning models (Isolation Forest and Random Forest) on Raspberry Pi 4 for hazard prediction. Established the LoRa communication pipeline (SX1278) for reliable wireless transmission, and expanded the RFID-based infrastructure for worker tracking across multiple zones.

Milestone 4 (Achieved in May 2026): Developed and finalized the dedicated supervisor application. Successfully deployed the complete "EDGE-ENABLED SMART MINE GUARDIAN SYSTEM" prototype, achieving real-time PPE violation detection via dual-camera architecture (e.g., entrances and Tunnel 1) and integrating all computer vision and RFID telemetry into a centralized, real-time monitoring dashboard.

Milestone 5 (Achieved in June 2026): Successfully engineered and deployed the portable, multi-modal worker device featuring an OLED display, auditory buzzer, and haptic vibration alerts, alongside an integrated SOS bidirectional transmission capability for enhanced individual occupational safety.

11. Research & Development Activities

The core R&D activities undertaken during this period involve:

- The formulation and development of an edge-enabled CPS architecture specifically tailored for the unique constraints of underground mine safety.
- Designing a robust multi-sensor data acquisition framework capable of operating accurately in harsh conditions.
- The implementation and optimization of TinyML-compatible anomaly detection algorithms tailored for localized edge processing.
- Developing multi-camera AI-driven computer vision solutions to autonomously verify PPE compliance across distributed physical locations.
- Engineering a real-time worker tracking system intricately coupled with portable worker alert devices and an automated emergency response mechanism.

- Integrating localized edge intelligence seamlessly with LoRa-enabled long-range communication networks.

12. Progress in Research

Research efforts during this quarter have been heavily focused on evaluating and optimizing intelligent algorithms for edge deployment. This entailed a comparative study of various edge AI algorithms to determine the most effective approach for hazard detection. Specifically, Isolation Forest and Random Forest models were rigorously evaluated for their efficacy and computational efficiency in identifying environmental anomalies.

Additionally, significant analytical effort was dedicated to the development and optimization of the multi-camera YOLO-based PPE detection system to ensure high accuracy within visually constrained environments like subterranean tunnels. The integration of RFID localization with AI-enabled monitoring systems was also a primary research focus, exploring novel data fusion techniques. The proposed CPS framework was subsequently validated through extensive prototype testing. The empirical results obtained thus far clearly demonstrate an improved hazard detection capability characterized by substantially reduced latency when compared to conventional, cloud-dependent monitoring paradigms.

13. Development Advances

Technological developments finalized during this quarter encompass the following functional components:

- A fully operational, edge-enabled multi-sensor monitoring prototype.
- An AI-based module dedicated exclusively to environmental hazard detection.
- A multi-camera AI-based monitoring and inference system for comprehensive PPE compliance.
- A portable, multi-modal worker device featuring OLED, buzzer, and haptic vibration alerts, alongside an integrated SOS transmission capability.

- An RFID-based infrastructure allowing for precise and continuous worker identification.
- A robust, low-power LoRa-based communication network.
- A dedicated supervisor application complementing the main mobile and web-based monitoring dashboard.
- An automated, low-latency emergency alert generation and dissemination module.

14. Impact & Significance

The system developed in this project represents a significant contribution to the evolution of next-generation, intelligent mine safety systems. Its primary impacts include:

- Enabling instantaneous, real-time hazard detection entirely independent of continuous cloud connectivity, thus mitigating the risks associated with network outages.
- Drastically reducing emergency response latency by processing critical data directly at the edge.
- Substantially improving overall worker safety through continuous, unbiased multi-camera AI-enabled PPE compliance monitoring.
- Enhancing individual worker security by providing dedicated, portable SOS devices with immediate, multi-modal feedback loops.
- Facilitating rapid and targeted emergency search and rescue responses through precise worker localization and supervisor mobile coordination.
- Delivering a scalable, highly modular, and cost-effective CPS framework that addresses the unique economic and physical challenges of underground mining environments.

15. Risk Management & Problem-Solving

Challenges Faced: The project encountered several significant challenges, including limited access to real-world, proprietary underground mining datasets. Furthermore, calibrating sensors across varying simulated environmental conditions proved complex. We also faced difficulties in ensuring communication reliability within confined and

heavily obstructed spaces, as well as maintaining optimal computer vision performance under inherently low-light and dusty conditions.

Mitigation Measures: To address these challenges proactively, we utilized established public benchmark datasets for the initial training, development, and validation of our AI models. We also conducted multiple iterative sensor calibration and validation experiments to ensure data accuracy and drift compensation. The adoption of a hybrid LoRa and MQTT communication architecture significantly bolstered transmission reliability and system redundancy. Furthermore, extensive hyperparameter tuning of the YOLO model, combined with advanced image preprocessing techniques (such as histogram equalization), effectively enhanced PPE detection accuracy even under suboptimal lighting.

Lessons Learned: The implementation process has highlighted several critical insights: deploying edge computing methodologies unequivocally reduces processing latency; sensor fusion is imperative for improving the overall reliability and reducing false positives in hazard detection; a hybrid communication approach is essential for maintaining robustness in network-deprived environments; and integrating portable, multi-modal hardware with AI-based monitoring can yield substantial, measurable improvements in the occupational safety of underground workers.

16. Technology Readiness & Roadmap

Current TRL: TRL-5 (The prototype has been successfully validated in a controlled laboratory environment).

Expected TRL by Project Completion: TRL-6 / TRL-7.

Prototype Status: The laboratory prototype is fully functional. The environmental sensing capabilities have been rigorously validated against baseline instruments, and all edge AI modules have been successfully integrated into the core system. The portable worker devices, supervisor application, and monitoring dashboard are currently operational, processing and visualizing data streams in real-time.

Future Roadmap: Moving forward, the project roadmap prioritizes conducting comprehensive field testing under simulated, realistic mine conditions to evaluate system robustness. We also plan to integrate additional specialized sensors to expand the monitoring parameter space. Future development will focus on incorporating predictive analytics capabilities for proactive hazard forecasting, and further optimizing system performance to handle extremely dusty and low-light environments. Concurrently, we will focus on finalizing our research for academic publication and exploring potential patent filings for the novel integrations developed.

17. Visual & Supporting Material

The following supplementary materials are compiled and attached alongside this report for comprehensive reference:

- Dashboard interface screenshots highlighting real-time environmental telemetry and AI-driven hazard analysis (e.g., Tunnel 1 High Humidity alerts).
- Worker tracking interface displaying individual personnel locations and exact "Last Seen" timestamps.
- Integrated Access & Tracking interface showcasing autonomous "Access Denied" logging and Activity Feeds based on multi-camera PPE non-compliance.
- Live YOLO inference captures demonstrating high-confidence bounding box detection of hard hats and safety vests.
- High-resolution photographs detailing the functional prototype and the portable worker alert devices.
- Close-up images of the hardware setup, including the ESP32, LoRa, OLED, and haptic motor integration.
- A comprehensive system architecture and data-flow diagram.
- Process flowcharts detailing the algorithmic logic of the edge nodes and the portable devices.
- A Gantt chart illustrating project timelines, milestones, and progress.
- Images documenting the experimental testing setups and calibration procedures.

Edge-Enabled Smart Mine Guardian System - Project Gantt Chart

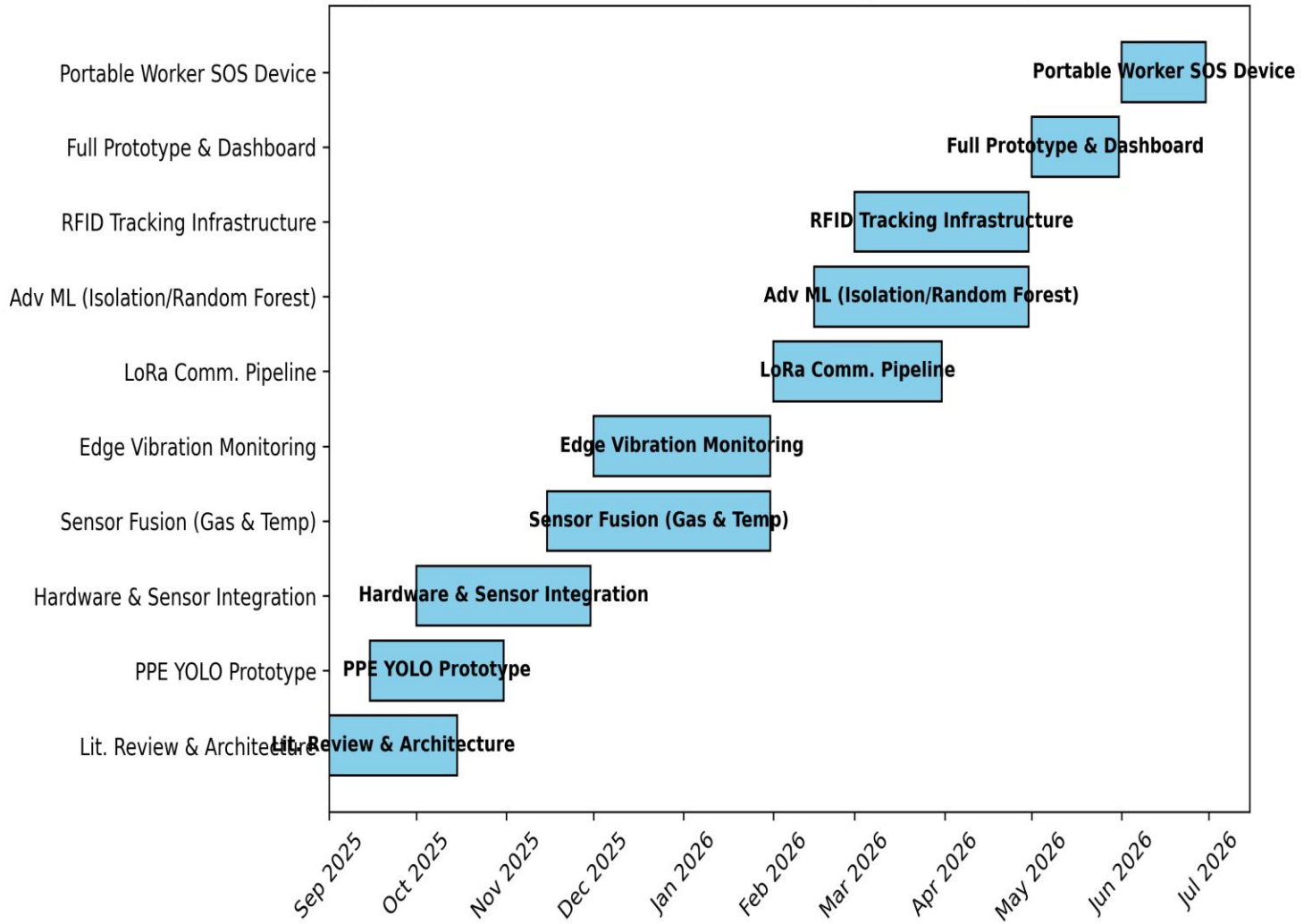


Figure 1: Project Gantt Chart Timeline (Sept 2025 - Jun 2026)

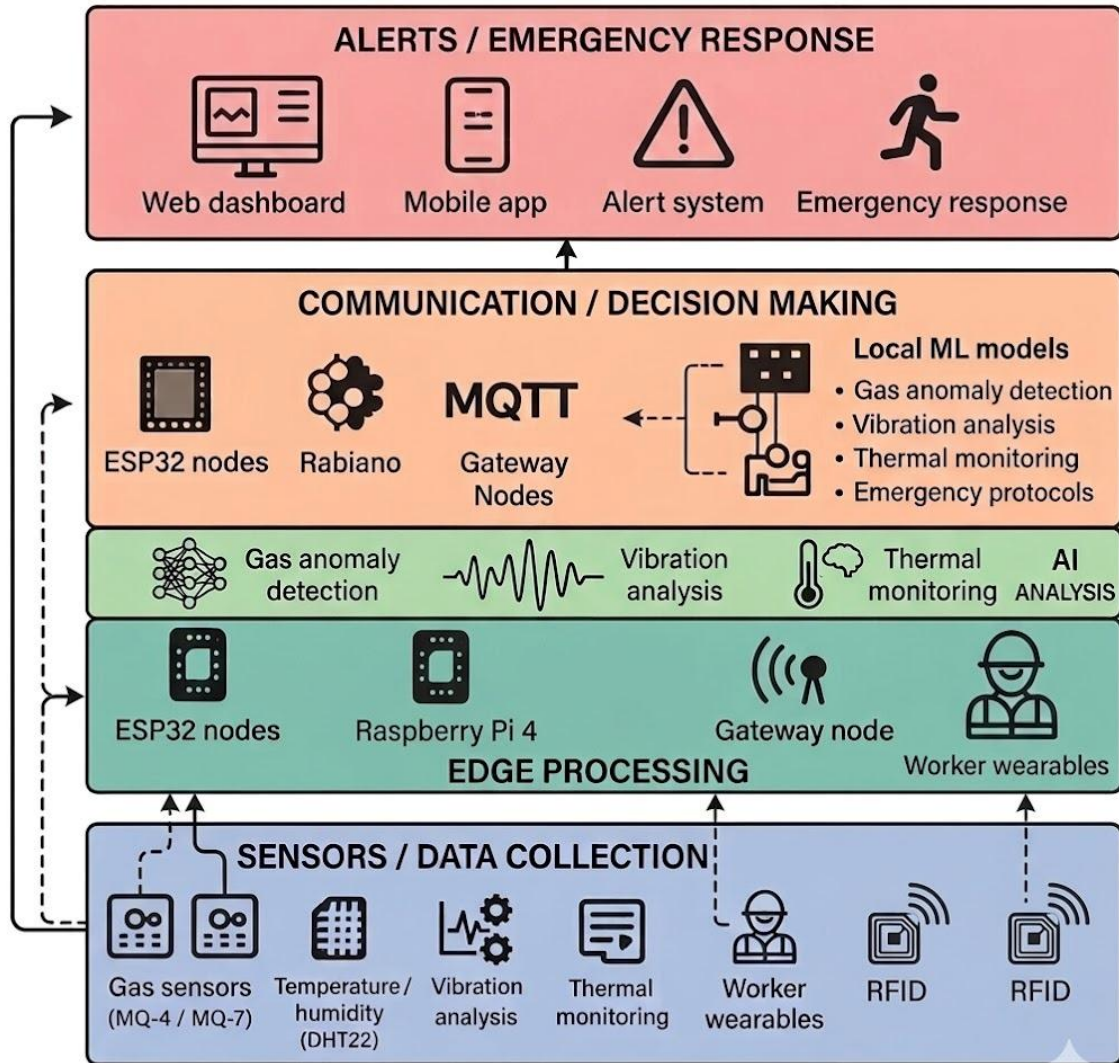


Figure 2: *System Architecture & Data-Flow Diagram*

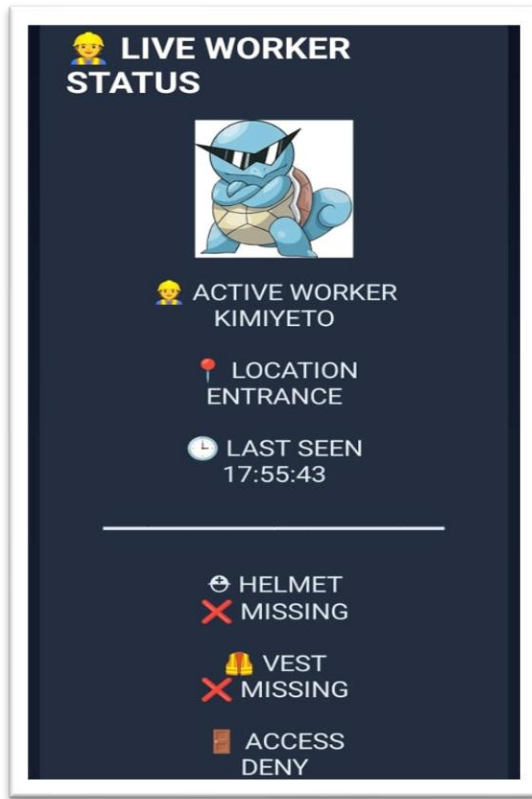


Fig 3: Live Worker Status Interface Showing PPE Violation and Access Denial

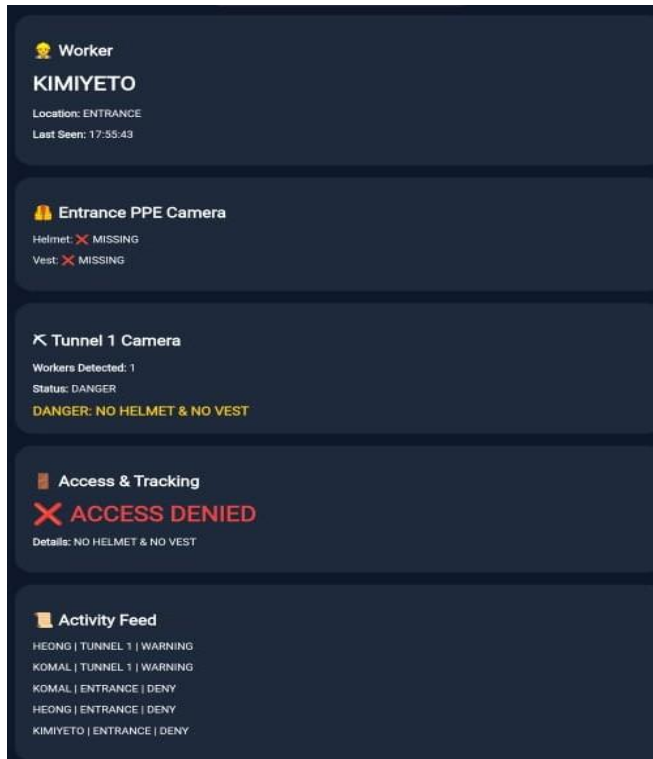


Fig 4: Activity Monitoring Dashboard During PPE Non-Compliance

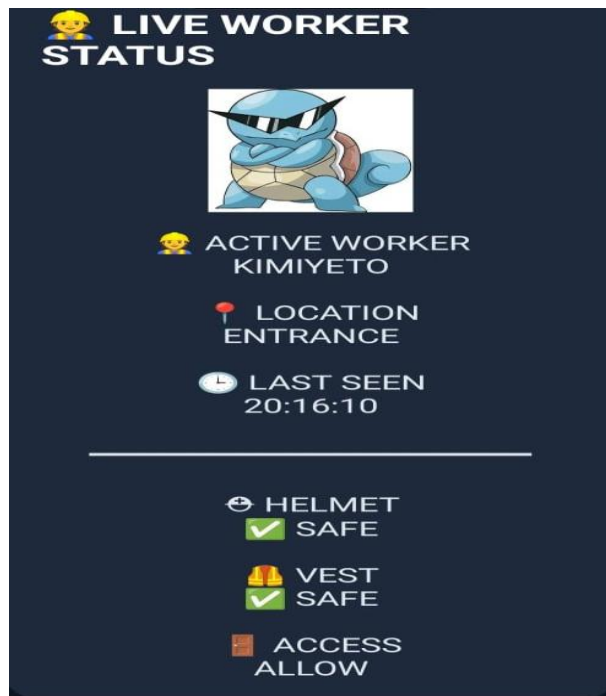


Fig 5: Live Worker Status Interface Showing PPE Compliance and Access Approval



Fig 6: PPE Detection of Safety Helmet and Reflective Vest



Fig 7: Real-Time Worker Monitoring



Fig 8: Hazard Alert Dashboard

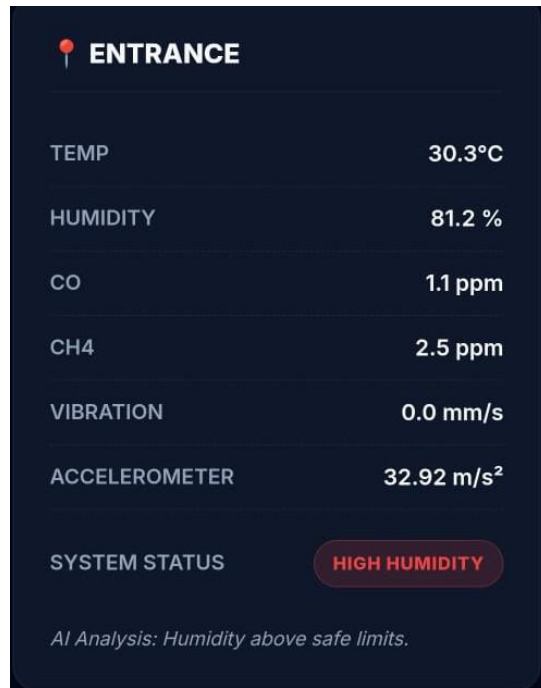


Fig 9: Entrance Environmental Monitoring

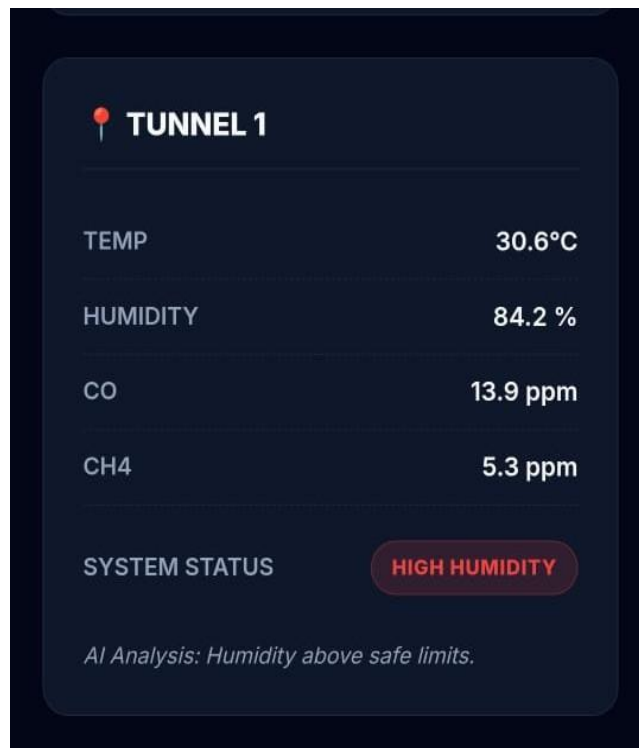


Fig 10: Tunnel 1 Environmental Monitoring

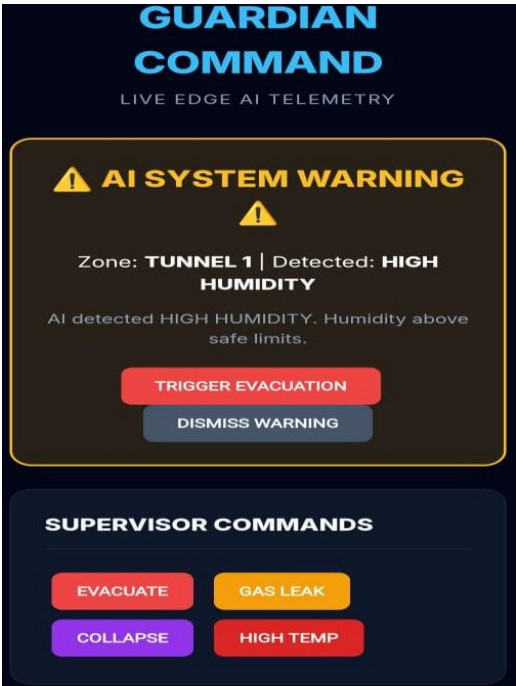


Fig 11: AI-Based Hazard Alerts and Supervisor Controls

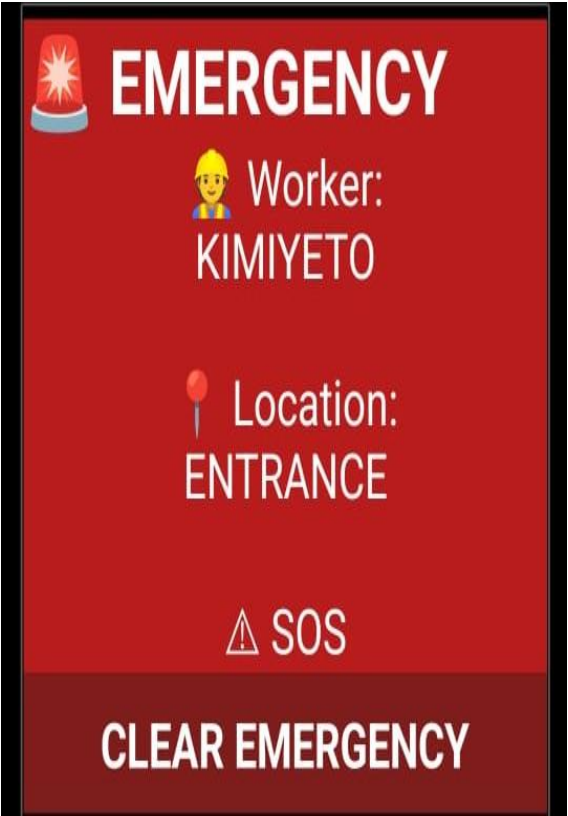


Fig 12: Emergency Response with Location

18. Deviation in Objective/Methodology (If Any)

Deviation: While the original project proposal outlined the deployment of a Jetson Nano for edge processing, the current implementation primarily utilizes a combination of Raspberry Pi and ESP32 microcontrollers. Additionally, the temperature anomaly model was initially trained using publicly available thermal datasets prior to its deployment and subsequent refinement with mine-specific sensor data.

Reason for Deviation: These deviations were necessitated primarily by the practical availability of hardware resources and the aforementioned limited access to real, proprietary underground mining datasets. The strategic shift to Raspberry Pi and ESP32 allowed us to maintain a strong focus on developing a highly cost-effective, easily deployable prototype suitable for rigorous academic validation, completely fulfilling the overarching objectives and scientific integrity of the project without compromise.

Acknowledgments:

I wish to place on record our deep sense of gratitude to our honorable guide, **Dr. Daijiry Narzary**, Assistant Professor, Department of Electronics and Instrumentation Engineering, National Institute of Technology Nagaland, for her invaluable guidance, insightful suggestions, continuous encouragement, and moral support throughout the course of this project. This work would not have been possible without her unwavering support and active involvement.

I would also like to express our sincere gratitude to **Dr. M. Prakash**, Head of the Department of Electronics and Instrumentation Engineering, National Institute of Technology Nagaland, for his constant encouragement and for providing the necessary facilities and support to successfully carry out this project.

I extend our heartfelt appreciation to **Dr. P. Chinnamuthu**, Faculty In-charge of **TEXMiN CPS Centre of Excellence (CoE)**, for his valuable guidance, encouragement, and continuous support throughout the project.

I also deeply thankful to **Prof. (Dr.) A. Elayaperumal**, Director of our institution, for his visionary leadership, constant motivation, and for providing an excellent academic and research environment that enabled us to accomplish this work successfully.

My sincere thanks are also due to **Mr. Karthikeyan**, Sub In-charge of **TEXMiN CPS Centre of Excellence (CoE)**, for his guidance, timely assistance, and valuable support during the execution of this project.

Finally, we express our heartfelt gratitude to our beloved parents for their unconditional love, unwavering support, and constant encouragement. We also offer our sincere thanks to the Almighty God for His blessings, strength, and guidance throughout this endeavor.

Date: 17-06-2026

Place: NIT NAGALAND

Signature of the Fellow

Faculty Mentor

(Signature with seal)

Head of the Department

(Signature with seal)